

Joseph Lavond

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EDUCATION

University of North Carolina

Ph.D. Statistics and Operations Research

M.S. Statistics and Data Science

Chapel Hill, NC

Aug. 2020 – May 2025

Aug. 2020 – May 2024

California Polytechnic State University

B.S. Statistics

San Luis Obispo

Sep. 2016 – Mar. 2020

EXPERIENCE

Data Scientist

June 2025 – Present

Red Ventures

Fort Mill, SC

- Built ReplayOPE, an offline policy evaluation platform for contextual bandits that surfaced stronger agent setups in 75%+ of experiments and generated automated recommendations for production decision systems
- Evolved ReplayOPE from research POC to internal ML product with FastAPI services, Databricks Apps, analyst UI, MCP integration, usage tracking, and CI/CD via GitHub Actions and Terraform
- Developed advanced evaluation methods including purged cross-validation, warm starts, delayed rewards, streaming execution, concurrent runs, eligibility checks, and rigorous policy comparison
- Improved DAPI offline RL infrastructure by fixing a 3+ year missing-values inference bug and productionizing CatBoost, ordinal LightGBM, and recency-based sample weighting / historical importance, with several agents observing +10 percentage-point revenue lift following adoption
- Drove adoption through analyst onboarding, product demos, searchable docs, and rollout of a vertical-wide skills marketplace for approved skill distribution

Graduate Data Science Intern

June 2024 – Aug. 2024

Red Ventures

Fort Mill, SC

- Implemented offline model-based Q-learning for website decisioning, demonstrating a typical 5% revenue-per-session lift and estimated \$5M annual revenue opportunity
- Built a historical-production-data simulation pipeline for Markov Decision Process experiments, increasing simulation quantity, complexity, and robustness

Graduate Research Intern

June 2023 – Aug. 2023

Cisco

San Jose, CA

- Developed scalable differentially private generative-model training methods and evaluated GAN mode-collapse mitigation techniques

Graduate Info Technology Intern

June 2022 – Aug. 2022

Elevance Health

Indianapolis, IN

- Built a PyTorch semi-supervised Bayesian anomaly-detection model on AWS and created a PySpark SQL pipeline for 60M-row claims data

PUBLICATIONS

- Joseph Lavond, Minhao Cheng, and Yao Li. Feddecay: Balancing model performance and rapid personalization in federated learning with learning rate scheduling. *Journal of Computation and Graph Statistics*, 2026. Accepted
- Joseph Lavond, Minhao Cheng, and Yao Li. Trusted aggregation (TAG): Backdoor defense in federated learning. *Transactions on Machine Learning Research*, 2024

SKILLS

ML & Decision Systems: Contextual Bandits, Offline Policy Evaluation (OPE), Reinforcement Learning, LightGBM, CatBoost, Sample Weighting, Bayesian Inference

Production ML & AI Infrastructure: FastAPI, Databricks Apps/Jobs, GitHub Actions, Terraform, CI/CD, Model Deployment, MCP, LLM Workflows, Experimentation Platforms

Programming, Data & Experimentation: Python, SQL, Pandas, NumPy, PySpark, SciPy, Feature Engineering, Data Pipelines, Statistical Testing, A/B Testing